

for agents with $b_i > 1/2$ and observing that disliking is a dominated strategy. Therefore, the right-wing island always shares, whereas the left-wing island has equilibrium cutoffs $(0, b_L^{**})$. Using the same approach as in part (a), it is enough to show that $U_L(\mathcal{S})$ increases following a decrease in divisiveness or polarization. Of course, for $b_i < 1/2$, we see that $\partial\pi_i/\partial p < 0$ and $\partial\pi_i/\partial q < 0$, and moreover, by Lemma A.3, decreasing polarization means that all agents on the left-wing island also have an increase in b_i . Thus, there is more sharing in the most-sharing equilibrium following a decrease in divisiveness or polarization. By strategic complementarity, the right-wing island remains at all-share, and sharing uniformly increases (and so does virality, naturally). ■

A.4 Proofs from Section 5

Proof of Theorem 3. Consider the complete sharing network where $\mathbf{P} = \mathbf{1}_{N \times N} - \mathbf{I}$. We claim that if the most-sharing equilibrium involves all agents choosing $\mathcal{S}$ or $\mathcal{I}$ (with probability 1) and agents never choosing $\mathcal{D}$ under this configuration, then this is the platform's profit-maximizing sharing network. By Lemma 1, all agents employ the same cutoffs $(b^*, b^{**}) = (0, b^{**})$ and $1 - H(b^{**})$ determines the proportion who share in the most-sharing equilibrium.

We focus on a modified social media game that only allows agents to ignore or share, which necessarily increases virality of content for any sharing network $\mathbf{P}'$ but does not increase the virality for the complete sharing network, by assumption. We show that for any other sharing network $\mathbf{P}'$, the largest fixed point (the most-sharing equilibrium), must necessarily be above $b^{**}\mathbf{1}$ (in the order $\preceq$). To do this, we consider the largest fixed point under $\mathbf{P}'$ (call this $\mathbf{b}^{**'}$), and use the same mathematical arguments as before (although we disregard the dislike cutoff). Let $\mathbf{B}'$ be the cutoff space where $\hat{\mathbf{b}}^{**}$ satisfies $\hat{\mathbf{b}}^{**} \preceq \mathbf{b}^{**'}$ and let the map $\varphi : \mathbf{B}' \rightarrow \mathbf{B}'$ map fixed cutoff strategies $\hat{\mathbf{b}}^{**}$ to best-response sharing cutoff strategies under the complete sharing network. It only remains to prove that φ indeed maps into $\mathbf{B}'$. To do this, let $U_j^c(\mathcal{S})$ be the utility from sharing under the complete network and $U_j'(\mathcal{S})$ as sharing under $\mathbf{P}'$, and note that $U_j^c(\mathcal{S}) - U_j'(\mathcal{S}) = \kappa \sum_{j=1}^N (1 - p'_{jj})(1 - H(\hat{b}_j^{**})) \geq 0$.

Thus, by Tarski's fixed-point theorem (Tarski (1955)), we obtain that $b^{**}\mathbf{1} \preceq \mathbf{b}^{**'}$. Finally, observe that this necessarily implies that $\mathbf{P}'$ is less viral, because for every prior realization and seed agent i^* , $\mathbf{S}_{i^*}$ is larger in the complete network than in any other sharing network, provided that $b^{**}\mathbf{1} \preceq \mathbf{b}^{**'}$ and $b^* = 0$ (no agent dislikes). By Proposition 1, (uniformly more) sharing is monotone in reliability, so there exists some r_P such that for $r > r_P$, the complete sharing network admits only shares and ignores, whereas when $r < r_P$, agents dislike with positive probability. When $r > r_P$, the network takes the form of part (ii) by setting $p_s = p_d = 1$.

Next, we consider the case where $r < r_P$, and so (b^*, b^{**}) are the cutoffs in the most-sharing equilibrium with a complete sharing network, but where $b^* > 0$. First, notice that there must exist an open interval where agents with priors $b_i \in (0, \bar{b})$ will *never* share, regardless of the sharing network $\mathbf{P}$. To see this, suppose that there exists some $\mathbf{P}'$ where *all* agents either share or ignore, so the equilibrium cutoffs are determined by $\mathbf{b}^{**'}$ (and $\mathbf{b}^{*'} = \mathbf{0}$). Using the reasoning as in the previous